

AI-Smart Campus System

Frontend Progress Report

Academic Performance Monitoring and AI-Based Recommendation System

Course	CSE 4204 — Mobile Computing Lab
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Team	CSE4204-8A-T07
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Submitted to

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1. Project Overview

The AI Smart Campus System is an intelligent academic support platform designed to assist students, faculty members, and administrators. The system provides secure authentication, AI-powered course recommendations, academic performance monitoring, risk alerts, and an AI chatbot using the Gemini API. The frontend was developed using React.js and Tailwind CSS and integrated with the Laravel backend APIs.

2. Frontend Technology Stack

Category	Technology
Frontend	React.js
Styling	Tailwind CSS
Backend	Laravel
Database	MySQL (XAMPP)
API Communication	Axios
AI Integration	Gemini API
Version Control	Git & GitHub

3. Frontend Project Structure

Description:

The frontend project follows a modular structure to improve scalability, maintainability, and code organization.

4. Completed User Interface

4.1 Landing / Home Page

The Home Page introduces the AI Smart Campus System and provides navigation to login, registration, and key system features.

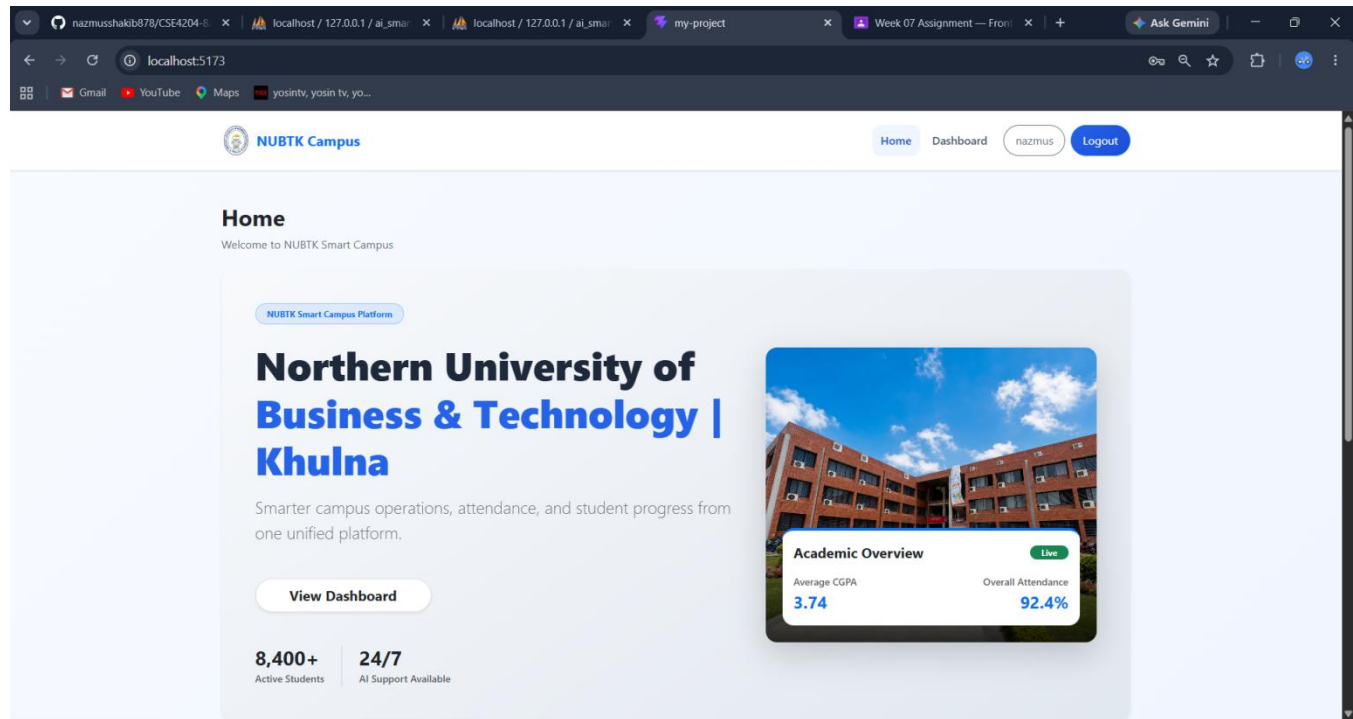


Figure 4.1: Home Page

4.2 Login Page

The Login Page allows registered users to securely access the system.

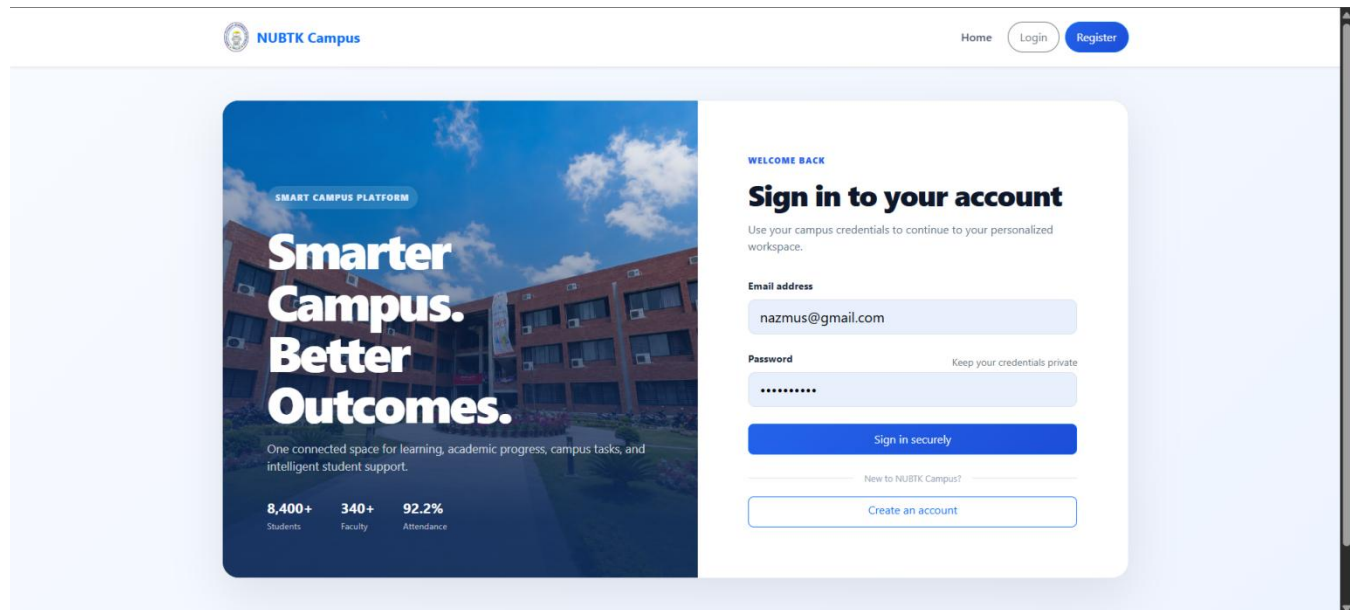
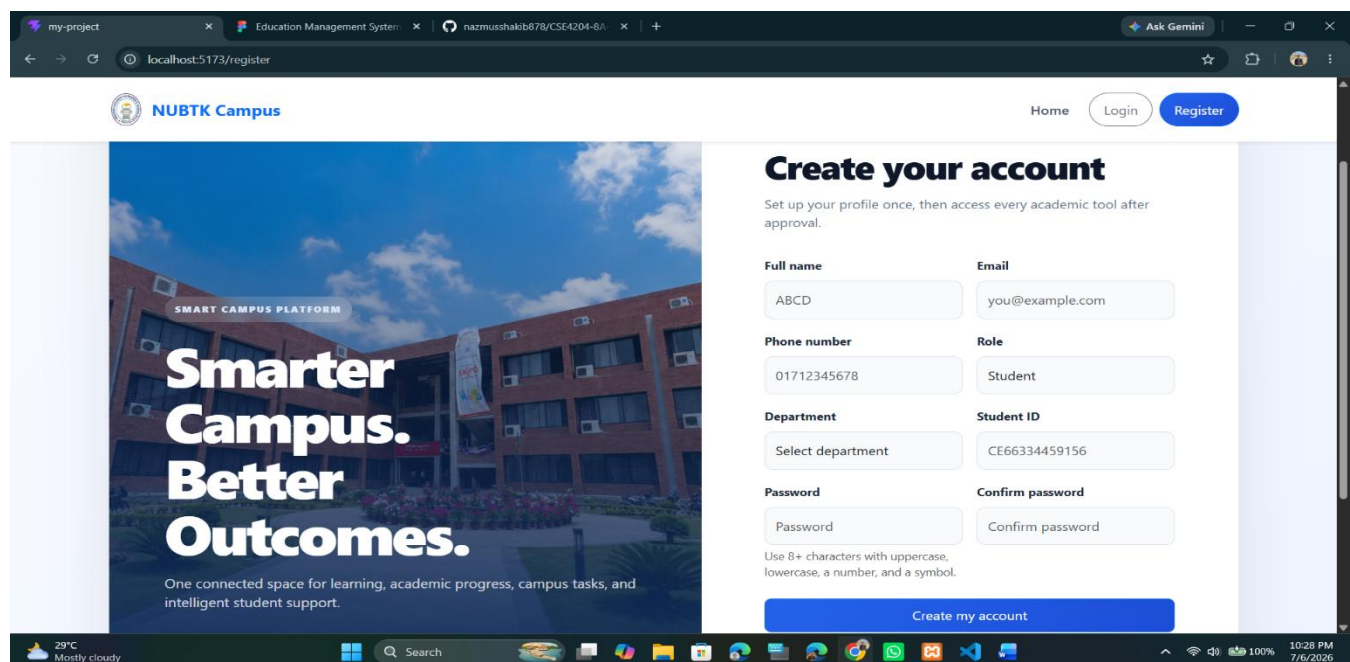


Figure 4.2: Login Page

4.3 Registration Page

The Registration Page enables new students and faculty members to create accounts.



4.4 Student Dashboard

The Student Dashboard displays attendance, quiz performance, course recommendations, AI chatbot access, and risk alerts.

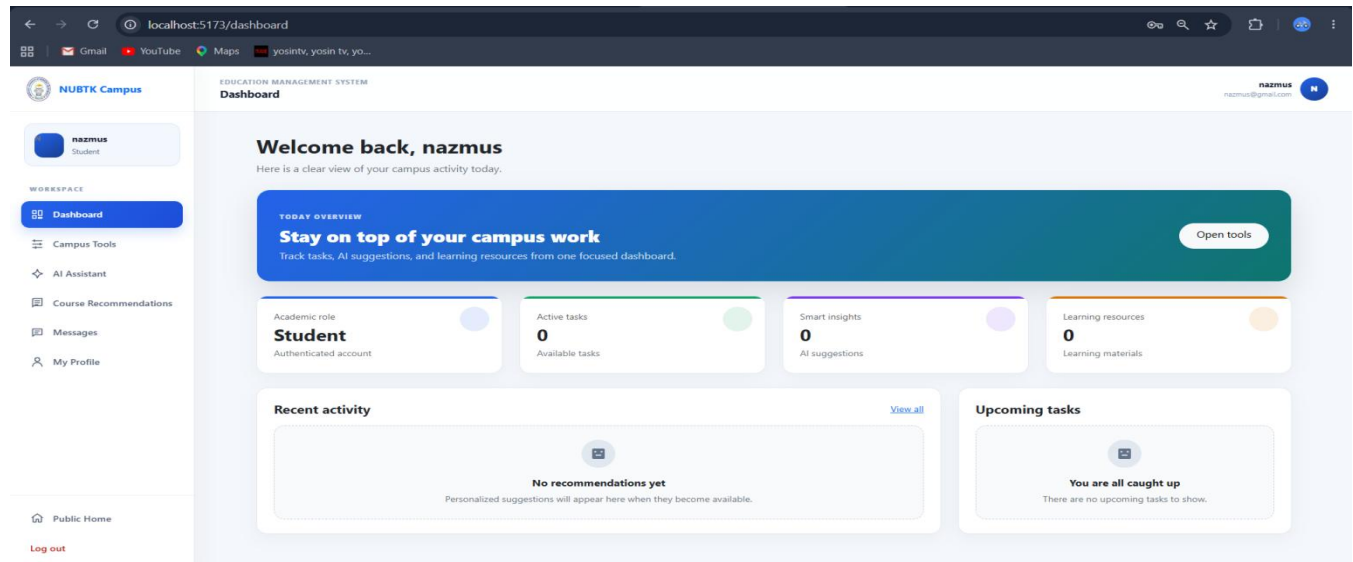


Figure 4.4: Student Dashboard

4.5 Faculty Dashboard

Faculty members can monitor attendance, academic performance, and identify at-risk students.

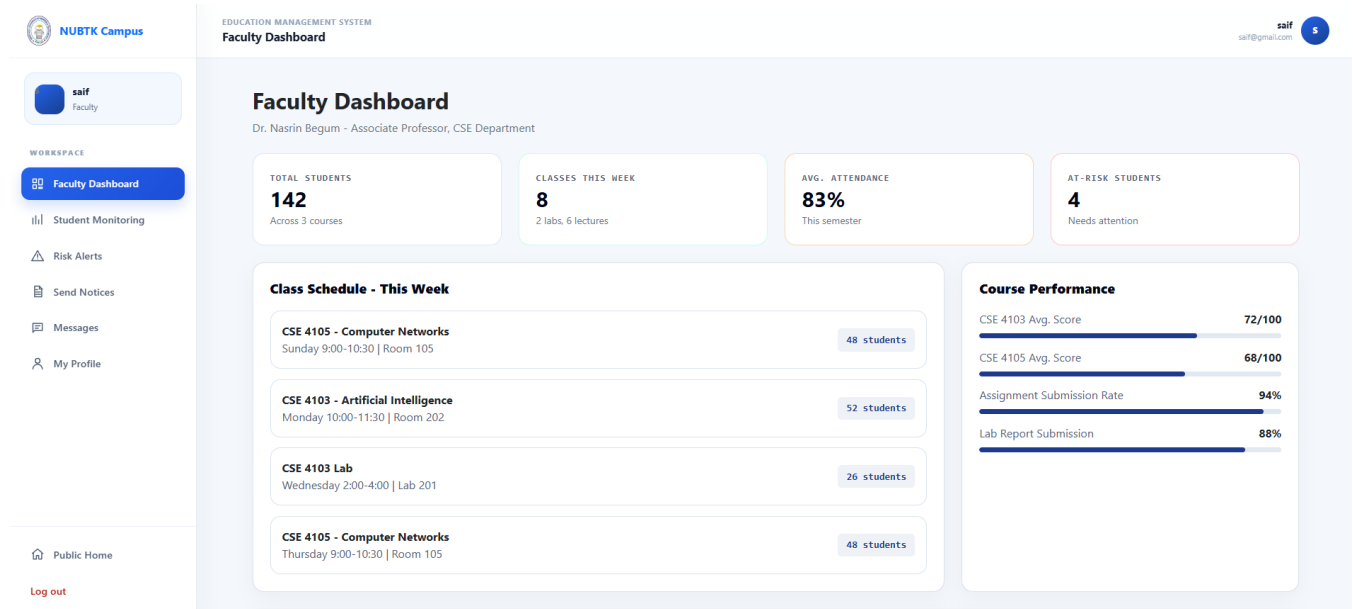


Figure 4.5: Faculty Dashboard

4.6 Admin Dashboard

The Admin Dashboard provides user management, approval, and overall system administration.

Figure 4.6: Admin Dashboard

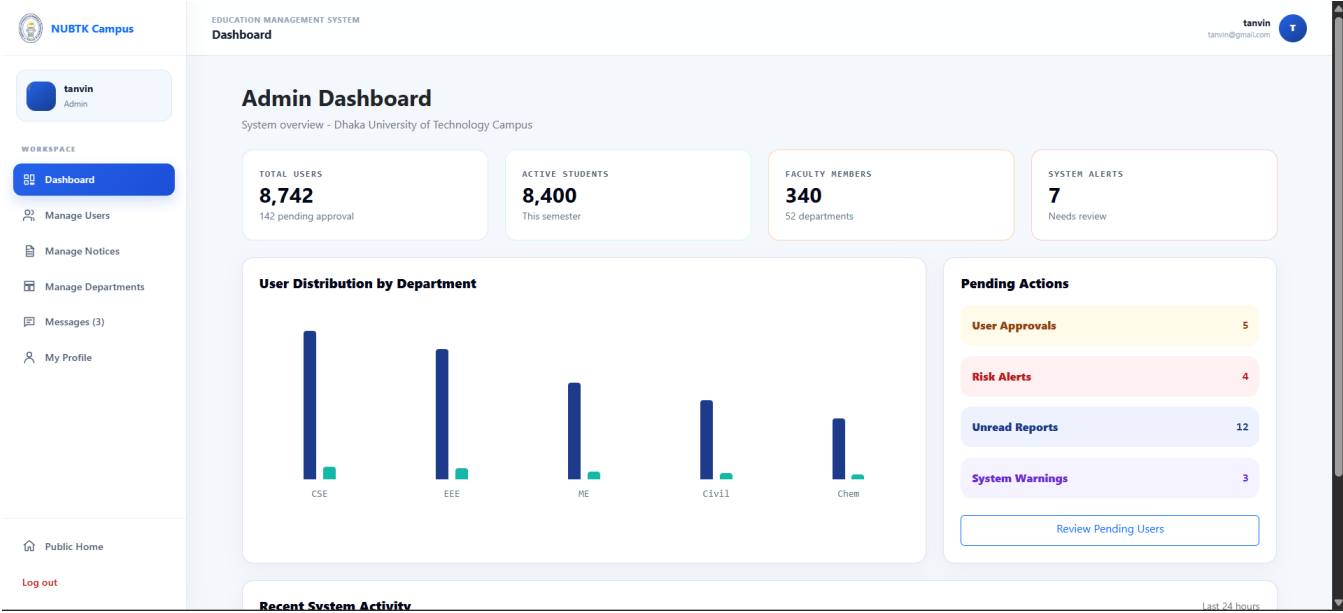


Figure 4.6: Admin Dashboard

4.7 User Profile Page

Users can view and manage their personal information.

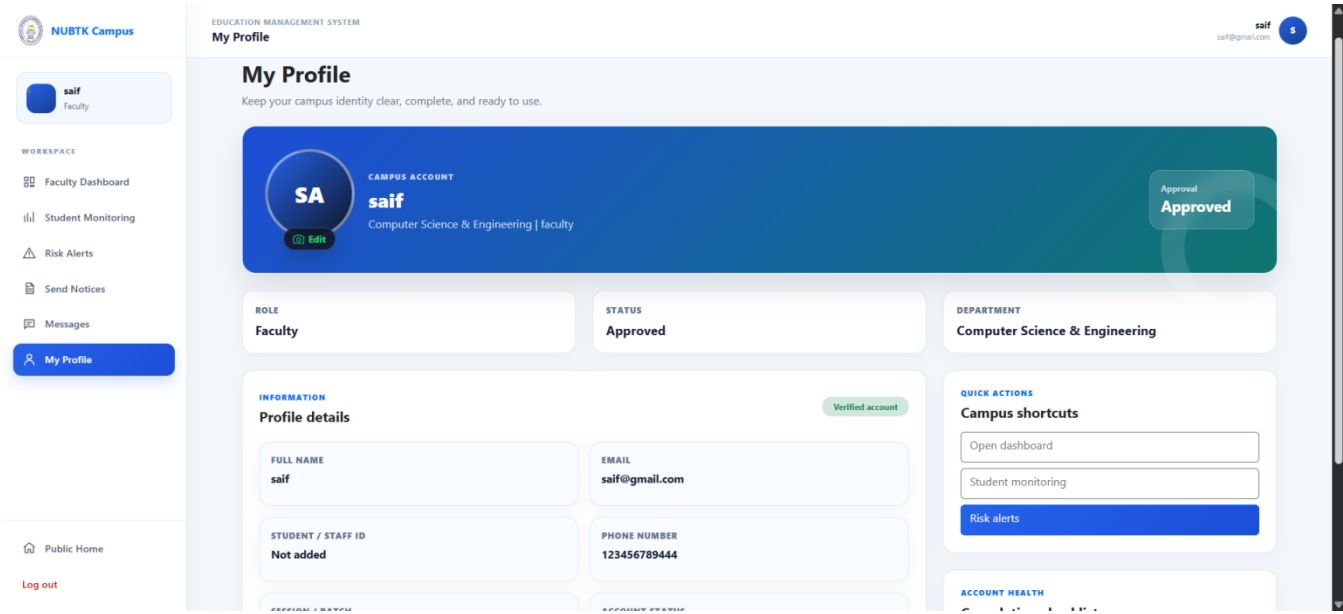
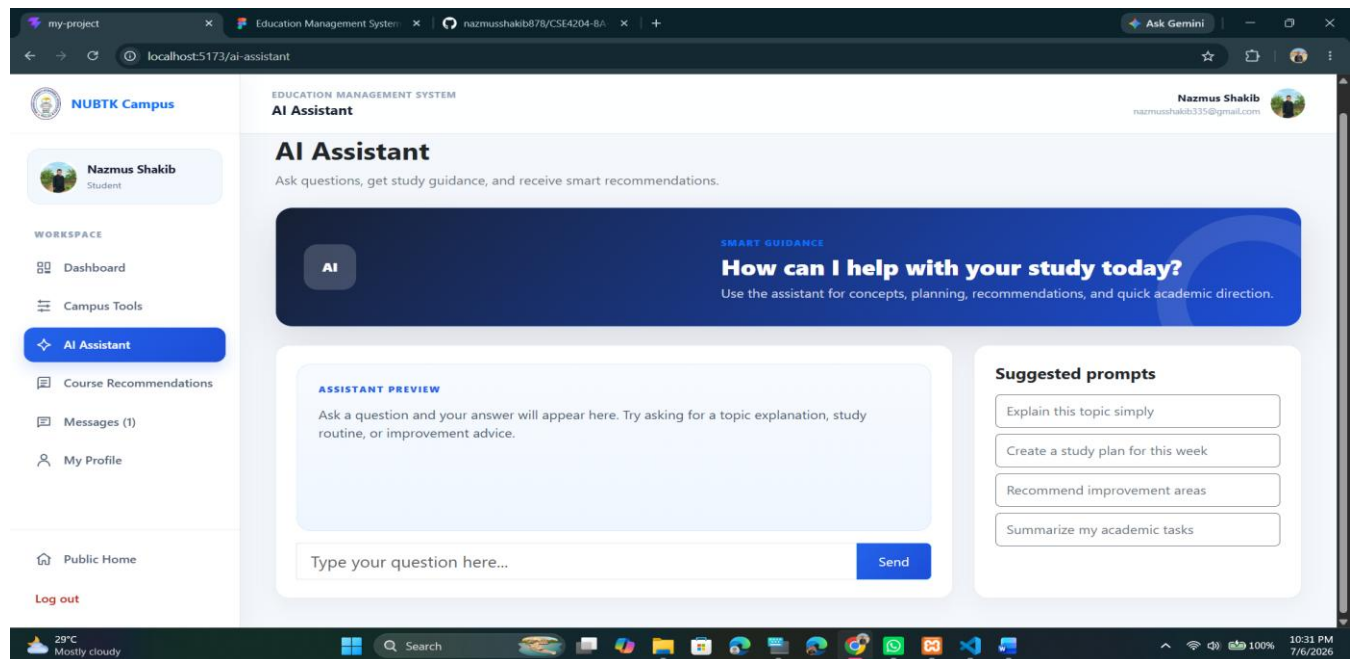


Figure 4.7: User Profile Page

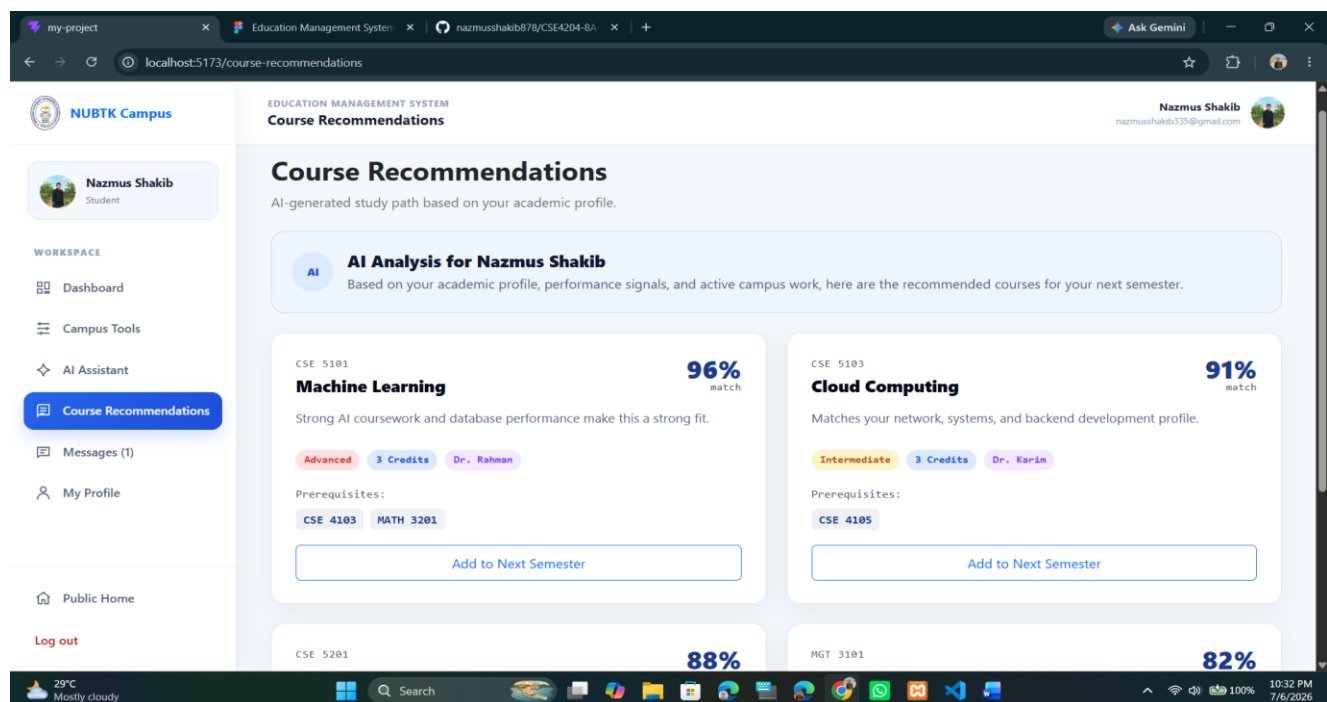
4.8 AI Interaction Page

The AI Chat page allows users to interact with the Gemini AI assistant for academic support.



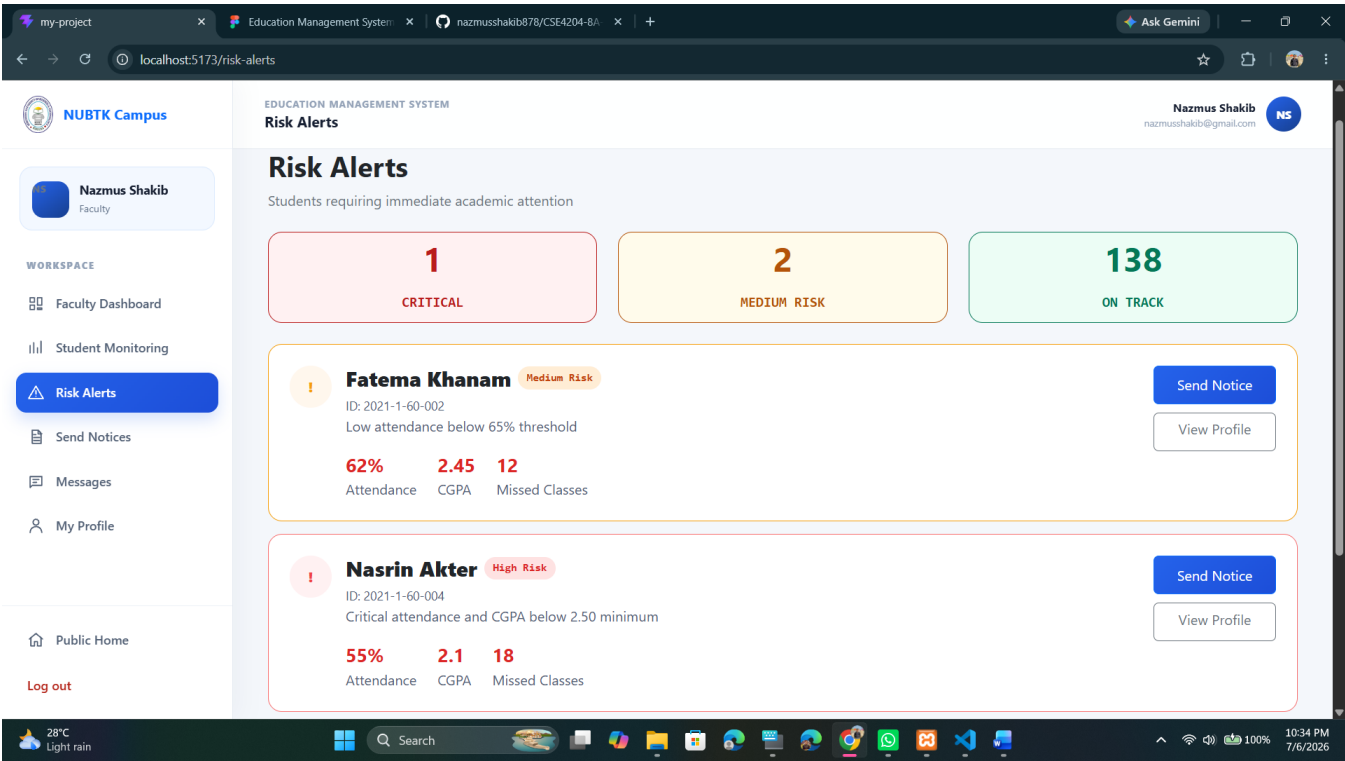
4.9 Course Recommendation Page

AI-generated course recommendations are displayed based on academic performance.



4.10 Risk Alert Page

Students receive alerts when attendance or academic performance falls below the required level.



5. Backend API Integration

The frontend communicates with Laravel backend APIs using Axios. All user information is loaded dynamically from the backend database.

API Table

API	Status
Registration API	Completed
Login API	Completed
Logout API	Completed
Profile API	Completed
Recommendation API	Completed
Learning Resources API	Completed

6. Authentication Flow

The authentication process uses Laravel Sanctum to provide secure login and protected access to authenticated users.

Authentication Flow

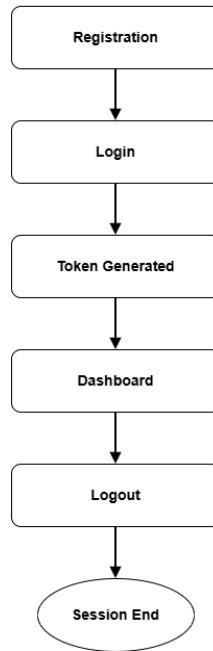


Figure 6.1: Authentication Flow

7. Form Validation

Client-side validation has been implemented to prevent invalid input and improve user experience.

- Validation Features
- Required Field Validation
- Email Validation
- Password Validation
- Error Messages

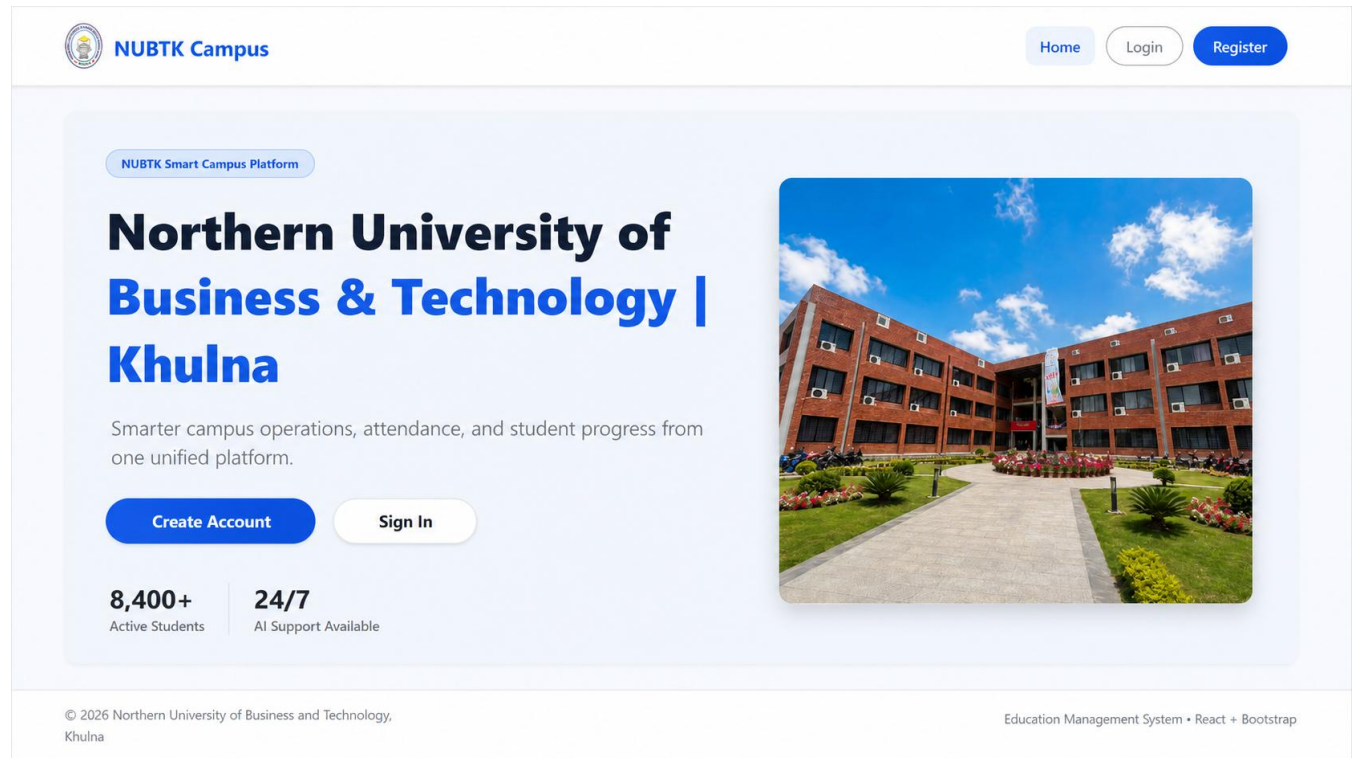
8. Responsive Design

Description:

The frontend interface has been optimized for desktop, tablet, and mobile devices.

Tested Devices

Desktop



Tablet**NUBTK Campus**[Home](#)[Login](#)[Register](#)

NUBTK Smart Campus Platform

Northern University of Business & Technology | Khulna

Smarter campus operations, attendance, and
student progress from one unified platform.

[Create Account](#)[Sign In](#)**8,400+**

Active Students

24/7

AI Support Available



Mobile

NUBTK Campus



NUBTK Smart Campus Platform

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Smarter campus operations, attendance,
and student progress from one unified
platform.

[Create Account](#)[Sign In](#)

8,400+
Active Students

24/7
AI Support Available



9. User Experience Improvements

Description

The application includes several user experience improvements:

- Loading Indicators
- Success Messages
- Error Messages
- Confirmation Dialogs
- Easy Navigation
- Responsive Layout

10. Current Development Progress

Completed Tasks

- ✓ Frontend Project Setup
- ✓ Home Page
- ✓ Login Page
- ✓ Registration Page
- ✓ Student Dashboard
- ✓ Faculty Dashboard
- ✓ Admin Dashboard
- ✓ User Profile
- ✓ AI Interaction Page

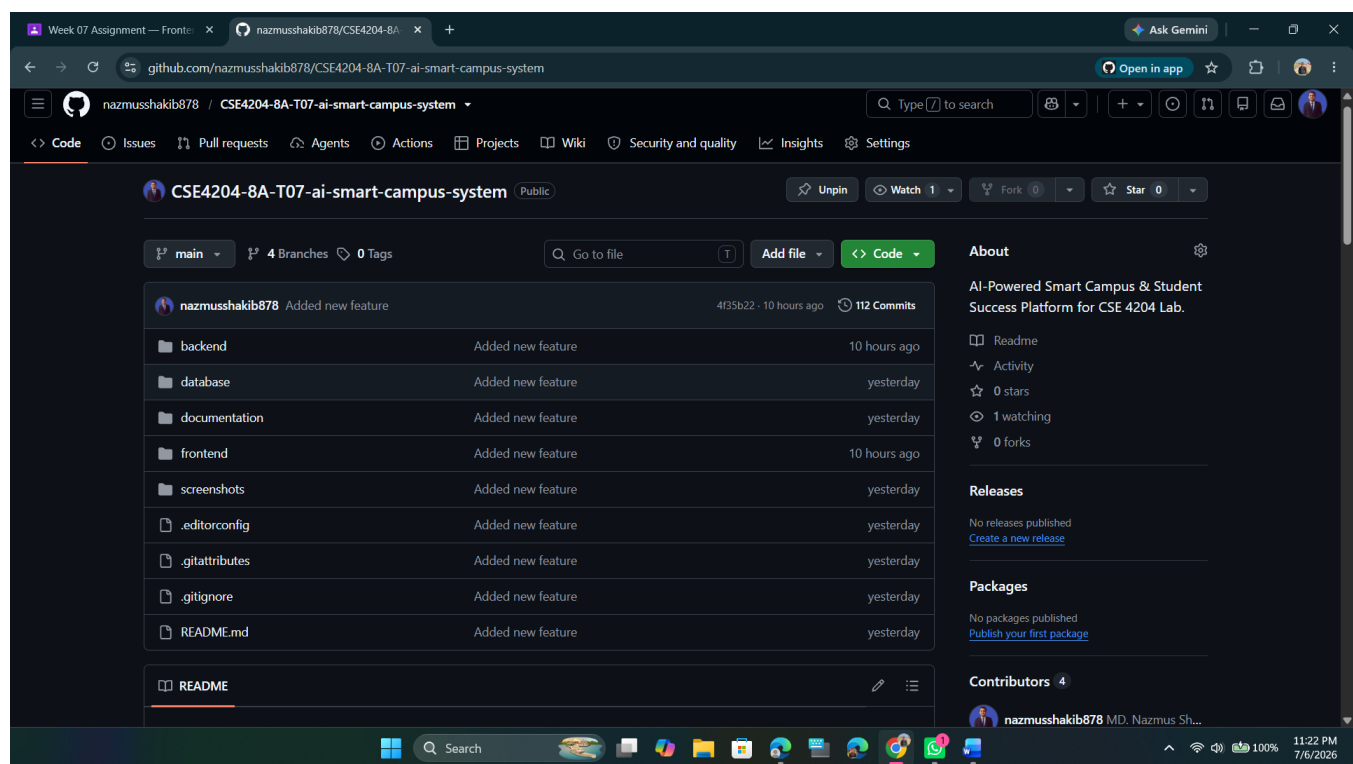
- ✓ Backend API Integration
- ✓ Authentication
- ✓ Responsive Design
- ✓ GitHub Repository Updated

11. GitHub Repository

Project Source Code (GitHub):

[CSE4204-8A-T07 AI Smart Campus System Repository](https://github.com/nazmusshakib878/CSE4204-8A-T07-ai-smart-campus-system)

11.1: GitHub Repository



12. Conclusion

The frontend development of the AI Smart Campus System has been successfully completed based on the approved UI/UX design. The application has been integrated with Laravel backend APIs, providing secure authentication, responsive user interfaces, and essential academic features for students, faculty members, and administrators. The project is well-organized and ready for the next phase of development.